Abdul Azeem Shaik

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EDUCATION

Georgia Institute of Technology
M.S. in Computer Science, Specialization : Machine Learning

BITS Pilani
B.E. in Electronics and Communication Engineering

PUBLICATION

"AI-based Market Intelligence Systems for Farmer Collectives" – ACM JCSS, funded by Google AI for Social Good; field-deployed engineer in rural India with IIT Delhi building custom solutions for farmers which had voice interactions, price predictions and weather forecasts.

EXPERIENCE

Georgia Tech | *AI Engineer / Researcher, Applied AI for Cybersecurity* Apr 2026 – Sep 2026

- Built multi-agentic AI systems with multi-turn communication for autonomous security incident triage, adding fault-tolerant quorum voting so agents stay reliable in autonomous Security Operation Centre (SOC) settings.
- Outperformed frontier LLMs on Microsoft's Excytin and OpenSec cyber threat investigation benchmarks; invited by faculty to TA *Modern Internet Research Methods: Agentic AI*, and recommended for AI security research with an industry defense partner.
- Built a RAG system with SecureBERT embeddings and agent knowledge bases from MITRE ATT&CK, D3FEND, CWE and Sigma rules; fine-tuned and deployed Qwen2.5 14B with LoRA for SecOps domain knowledge; built an XGBoost + HDBSCAN classifier to detect malicious injection in IT logs.
- Evaluated prompt injection resistance through adversarial test pipelines based on Wazuh schemas for load testing

Liberdus | *Software Engineer, Product & Scalable Systems* Jan 2026 - Present

- Brought down transaction time to finality by 85% by fixing deadlocks in the signing layer; cut historical data sync latency by 63% by adding gossip protocols to the sync pipeline; reduced heap memory usage by 50% via RPC cache optimization, stopping OOM crashes; fixed database contention between data collectors and writer services using BetterSQLite.
- Primary security owner: audited the execution layer and fixed critical CVEs; hardened REST APIs against DDoS across the signing infrastructure and RPC clients; used agentic AI workflows to speed up incident response for issues in production (MTTR).
- Built a quorum-based validator mechanism for reliable client-side communication alongside configuration enforcement across decentralized nodes; wrote cryptographic libraries in Rust for identity verification and threshold signatures.
- Built user-facing payment products end to end, including the Liberdus Bridge transaction dashboard and payment UI in React and Node.js

Shardeum *Raised $24M in funding at a $248.9M valuation. Early technical hire, helped launch mainnet and testnet.*

Senior Software Engineer, Distributed Systems Jul 2025 – Dec 2025

- Contributed core features to a distributed platform scaling to 80M+ transactions and 150K+ registrations; redesigned storage architecture and built multi-shard state sync for network-wide consistency; built payment APIs at ∼30 ms P99 latency, an RPC reverse proxy with node discovery and load balancing, and a checkpoint-aware patching system for fault-tolerant rollbacks and crash recovery.
- Built the cross-network payment platform and UI in React, Node.js and TypeScript, and the client-facing network health dashboard in Vue.js and Nuxt.js; set up production monitoring and alerting on AWS (CloudWatch, Lambda, Jenkins CI/CD); built an MCP server enabling JSON-RPC communication between autonomous agents for payment workflows and AI-assisted financial analysis tooling
- Led the redesign for authorization verification of incoming transactions; helped leadership model the network's governance & incentive plan.

Software Engineer, Distributed Systems Aug 2022 – Jul 2025

- Handling 10,000 req/sec, designed and built a horizontally scalable node-monitoring service for decentralized communication networks, tracking node health and network state across a distributed peer set without a central bottleneck.
- Improved ECDSA address recovery latency 14× (10 ms → 700 μs) by rewriting the core logic in Rust; improved validator networking latency 34% (82 ms → 54 ms) in the TCP-based async library; cut signing latency 29% by moving the crypto library to Rust-based `libsodium`; built JSON-RPC APIs with GraphQL data services, reducing response delay by 78%.
- Built a custom serialization library for faster, secure gossip-based P2P; upgraded the data distribution protocol with WebSocket and RPC support
- Found and fixed critical bounty bugs covering the Proof of Quorum consensus algorithm, DDoS, MITM and replay attacks, preventing double-spend exploits and unauthorized fund transfers on the live network; improved lost-node detection and added penalties for malicious activity

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust
Frameworks & Tools: Node.js, FastAPI, React, MySQL, SQLite, Docker, Git, AWS Cloud Infrastructure
AI & Machine Learning: AWS Certified AI Practitioner ; Vector DBs (Qdrant), RAG, PyTorch, HuggingFace, LangGraph, MCP

OPEN SOURCE

- **Red Hat** : Hardened Red Hat's MAPT provisioning engine against malformed AWS and Azure Kubernetes secret inputs
- **Farcaster** : Engineered chain sync observability in Snapchain via `BLOCK_CONFIRMED` events and filtered GetEvents queries